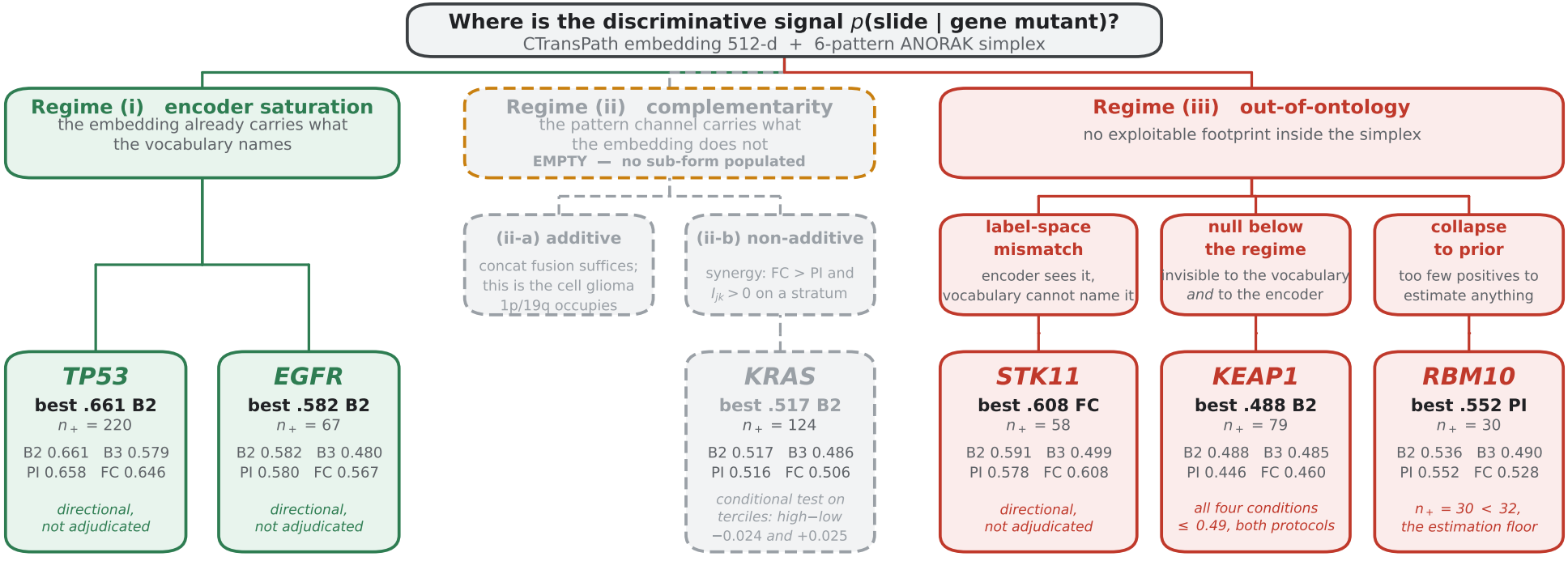
What kind of problem is morphology → mutation prediction for each gene?

Three regimes, read as the atoms of the partial information decomposition of the pattern and embedding channels

Nested cross-validation, 444 patients, five seeds, patient-preserved folds. No regime assignment is adjudicated.



Cohort verdict (Hosmer−Lemeshow gate, AUROC threshold 0.70)

Conclusive (AUROC ≥ 0.70)

none

the box was populated before the cohort was corrected

Inconclusive (AUROC < 0.70)

TP53 (.661) EGFR (.582) STK11 (.608) RBM10 (.552) KRAS (.517) KEAP1 (.488)

all six targets, under both split protocols

Values are pooled out-of-fold AUROC, mean over five seeds (tab_mutation_luad, patient-preserved panel). B2 = ABMIL on embeddings; B3 = ABMIL on patterns; PI = concat fusion; FC = 2-additive Choquet. The only contrasts that resolve are B3−B2 on TP53, EGFR and STK11; no fusion contrast resolves on any gene, and no equivalence is established at the ±0.02 margin the regime reads require, so every assignment above is directional.